

Cell Painting Shortcut Audit: does morphology beat a cell-count baseline, once we correct for testing hundreds of endpoints?

Competition: AI for Life Science Challenge (5th Pazhou Algorithm Competition) · **Category:** Model & Algorithm **Team:** Vrushti (biology and data lead: labels, dataset QA, biological interpretation) and Dhairya (ML and statistics lead: models, calibration, multiple testing, demo), a biology × machine learning pairing **Code:** <https://github.com/dhairya-pandya/beyond-the-count> · **Demo:** <https://beyond-the-count1.streamlit.app/> · **Video:** <

count1.streamlit.app/ · **Video:** <

All numbers are regenerated from `results/` (see §7). The video link in `<<...>>` is filled in once the video is published.

1. Problem definition

Image-based phenotypic profiling (Cell Painting) is being adopted as a scalable, human-relevant new approach methodology (NAM) for toxicology. Models trained on such profiles can score well for the wrong reason: for cytotoxicity-related readouts, a model that only *counts surviving cells* can match a sophisticated morphology model, so an apparent "morphology signal" may be a cell-count shortcut. The source study (Ewald et al., 2026) compares morphology to a cell-count baseline, but, over ~400 external endpoints, does not (i) correct for the number of endpoints tested, (ii) separate endpoints with too few positives to support any conclusion, or (iii) test whether morphology's advantage clusters in a coherent biological family.

Contribution. A reusable, dataset-agnostic *audit* that answers, per endpoint: (1) does the full morphological profile beat a cell-count-only model after Benjamini–Hochberg FDR control across all tested endpoints? (2) is there enough data to say — otherwise the endpoint is reported as *indeterminate*; (3) where morphology wins, is the advantage concentrated in an assay/target family, or arbitrary?

Scope and non-claims. The data are a 2D primary-human-hepatocyte plate assay, not an organ-on-a-chip. We position the work as a validation-methodology contribution relevant to organ-on-chip toxicology, whose readouts face the same shortcut risk; we do not claim chip results. No raw-image modeling (precomputed profiles only), no new wet-lab data, a single 44 h time point, donor variability not modeled.

2. Data sources and licenses

Item	Source	License
Well-level profiles (CellProfiler, CP-CNN, DINOv2), metadata, image index	Zenodo 10.5281/zenodo.17067683	CC-BY 4.0
ToxCast/Tox21 binary hit-calls, endpoint annotations (target family, assay design), SI tables (LDH/MT/cell-count hits and points of departure)	github.com/jessica-ewald/2024_09_09_Axiom_OASIS	BSD-3-Clause
Source paper	Ewald et al., <i>Cell Systems</i> 17(5):101566 (2026), doi:10.1016/j.cels.2026.101566	—

Design of the source experiment: 1,085 compounds (pharmaceuticals, pesticides, industrial chemicals), 8 concentrations $\times$ 2 replicates, primary human hepatocytes, 384-well, 44 h exposure; LDH, metabolic-activity (MT) and Cell Painting (6 dyes, 5 channels) readouts with cell count extracted from the images. We use only the published, precomputed profiles; raw images (Cell Painting Gallery, ~650 TB bucket) are not downloaded.

2.1 Dataset statistics

quantity	value
wells / plates / DMSO control wells	21,456 / 65 / 4,849
compounds with an OASIS id (profiled, modeled)	967 (966 for DINOv2) of 1,086 compound names; the rest are blinded
concentrations per compound	8 (one compound 7), 2 replicates, spanning 13–16 plates from two production batches
raw $\rightarrow$ retained features after variance and correlation (	r
endpoints	292 cell-based + 72 cell-free + 38 cytotoxicity (ToxCast/Tox21) + MT, LDH, cell_count (native) = 405
profiled compounds with a label per endpoint, median (min–max)	cell-based 202 (10–652) $\cdot$ cell-free 20 (7–131) $\cdot$ cytotoxicity 229 (13–667)
actives per endpoint among profiled compounds, median (10th–90th percentile)	cell-based 15 (5–39) $\cdot$ cell-free 8 (4–23) $\cdot$ cytotoxicity 69 (5–132)
endpoints with ≥ 15 actives and ≥ 15 non-hits among modeled compounds	181 of 404 (cell-based 147 / 292, cytotoxicity 24 / 38, cell-free 8 / 72, native 2)

Label provenance: the Zenodo record contains profiles and metadata only; outcome labels were located in the authors' public analysis repository (`1_snakemake/inputs/annotations/*_binary.parquet`, `2_downstream_analysis/compiled_results/SI_tables/`). We therefore did not need to re-derive hit-calls from EPA invitrodb. The repository's cytotoxicity file contains 38 endpoint columns (the paper text refers to 48); we use what is shipped.

3. Methods

3.1 Profile preprocessing (re-implemented from the authors' pipeline)

Per plate, features are centred and scaled by the median and MAD of that plate's DMSO controls; features that are non-finite, or have zero MAD / $|\text{MAD}/\text{median}| \leq 10^{-3}$ in any plate, are dropped; a greedy correlation filter ($|r| > 0.9$) removes redundant features. Per-compound profiles are obtained by averaging treated wells under three aggregation rules taken from the source repository: `all` (all wells), `allpod` (wells above the compound's Cell Painting point of departure, POD), `allpodcc` (wells above the POD and below the cell-count POD, with the source's fallbacks). PODs are taken from the published SI tables. The unit of analysis is therefore the compound: labels are per compound, each compound contributes one row, and the default `allpod` rule keeps only wells at or above the compound's point of departure, so low concentrations at which nothing happens do not dilute the profile or inherit an "active" label (§4.10 reports the other two rules).

3.2 Models

For every endpoint, three XGBoost classifiers (150 trees, learning rate 0.05, `scale_pos_weight = neg/pos` — identical to the source classifier) are trained on identical folds:

- **full** — the morphology profile (CellProfiler primary; CP-CNN and DINOv2 as robustness checks);
- **scalar_cc** — the source paper's baseline: the mean cell count (one number);
- **strong_cc** — a deliberately *stronger* shortcut model that still sees **only cell-count information**: the scalar count, cell count at each of the 8 concentrations ($\div$ plate DMSO median), their minimum and mean, and the cell-count POD.

Hyperparameters are fixed to the source paper's values, so no tuning ever sees a held-out fold; two further models answer design questions raised in review (§3.8).

The strong baseline is the primary comparator: a verdict against a one-number baseline can be an artefact of that baseline being weak. The technical-covariate baseline in the original plan (plate, well position, batch) is not applicable at compound level, because each compound's profile averages wells from 13–16 plates.

3.3 Cross-validation and effect estimate

Stratified, compound-grouped, shuffled 5-fold CV (folds reduced to $\min(5, n_{\text{pos}}, n_{\text{neg}})$ for sparse endpoints), repeated 3× with different fold seeds; out-of-fold probabilities are averaged over repeats. $\Delta\text{AUROC} = \text{AUROC}(\text{full}) - \text{AUROC}(\text{baseline})$ on pooled out-of-fold predictions. A paired bootstrap over compounds (1,000 resamples) gives a 95% interval and a standard error SE; the raw one-sided p-value is $p = (1 + \#\{\Delta \leq 0\})/(B + 1)$. *This captures evaluation-sample* variability only* — not the variability of the fitted models — which turns out to matter (§3.4).

3.4 Auditing the audit: label-permutation null and calibrated p-values

We re-ran the identical pipeline on hundreds of endpoints whose labels were randomly permuted across compounds (true $\Delta = 0$ by construction; `scripts/08_null_control.py`). Under this null the z-score $z = \Delta/\text{SE}$ has a standard deviation clearly above 1, so raw bootstrap p-values are anti-conservative. Following Efron's empirical-null idea we estimate sd_0 from the powered permutation runs and use the

calibrated one-sided p-value $p_{\text{cal}} = P(Z > z/sd_0)$. The estimate is feature-set specific (§4.4); split-half validation (sd_0 from the first runs, error rates on held-out runs) checks that the calibration holds. Raw bootstrap p-values and verdicts are retained alongside (`bootstrap_p`, `verdict_uncalibrated`).

3.5 Power check, multiple-testing correction, verdicts

An endpoint is *powered* if it has ≥ 15 actives and ≥ 15 inactives among profiled compounds (pre-specified heuristic; sensitivity in §4.10 — it was not tuned after seeing results). Benjamini–Hochberg FDR is applied to the calibrated p-values across **powered** endpoints only. Verdicts: `morphology_advantage` (powered, $\Delta > 0$, $q < 0.05$), `no_advantage` (powered, otherwise; displayed as *no detectable advantage*), `indeterminate` (under-powered; excluded from the FDR pool and from the enrichment test, never reported as a finding either way). The `cell_count` native endpoint, whose label is defined from the cell-count curve, is a positive control (the strong baseline must reach AUROC ≈ 1) and is outside the pool. BH assumes independence or positive dependence; endpoints share compounds and are correlated, which we flag as a limitation.

3.5b Reading `no_advantage`, correlated tests and precision

`no_advantage` is displayed as **no detectable advantage**: with enough data to compare, no difference survived correction. That is absence of evidence, not an equivalence result, and no equivalence margin is claimed. How informative such a verdict is depends on the width of the uncertainty around the AUROC difference, not on the 15/15 count, so every endpoint also gets a **minimum detectable effect** (MDE): from the paired-bootstrap interval we recover SE, widen it by the permutation-null sd_0 , and report $(z_{0.95} + z_{0.80}) \cdot sd_0 \cdot SE$, the smallest true AUROC difference a one-sided $\alpha = 0.05$ test would find with 80% power (`scripts/23_detectable_effect.py`). BH is valid under independence and under positive dependence, which describes endpoints sharing compounds and assays; the effective number of independent tests is smaller than the nominal count, so the correction is conservative rather than anti-conservative.

3.6 Calibration

On out-of-fold probabilities of the full model: Brier score, Brier skill versus the prevalence predictor, expected calibration error (10 bins), calibration slope of a logistic recalibration, and the Brier score after cross-validated Platt recalibration. Because training re-weights classes, raw probabilities are scores rather than frequencies; the demo shows this explicitly.

3.7 Biological enrichment

For each grouping (assay/target family, assay design type, cell line, tissue, category), a one-sided Fisher exact test asks whether the group is over-represented among `morphology_advantage` endpoints relative to the other powered endpoints (min group size 5; BH within each grouping). Target families come from EPA's own `intended_target_family` annotation, so no per-compound annotation is required. Endpoints from one assay share compounds, cell system and readout, so they are not independent observations. Each group therefore also gets an **assay-level p-value** (`p_cluster`, BH-adjusted as `q_cluster`): endpoints are clustered by the first two tokens of their id (106 clusters, e.g. `BSK_SAg`, `T0X21_ERa`), each cluster is assigned to the level most of its endpoints belong to, and the same number of clusters is drawn at random as the null 'in' set, so a group that is a single assay cannot reach a p-value below about $1/(\text{number of clusters})$. Null results are reported as such.

3.8 Nested model and a regularised comparator

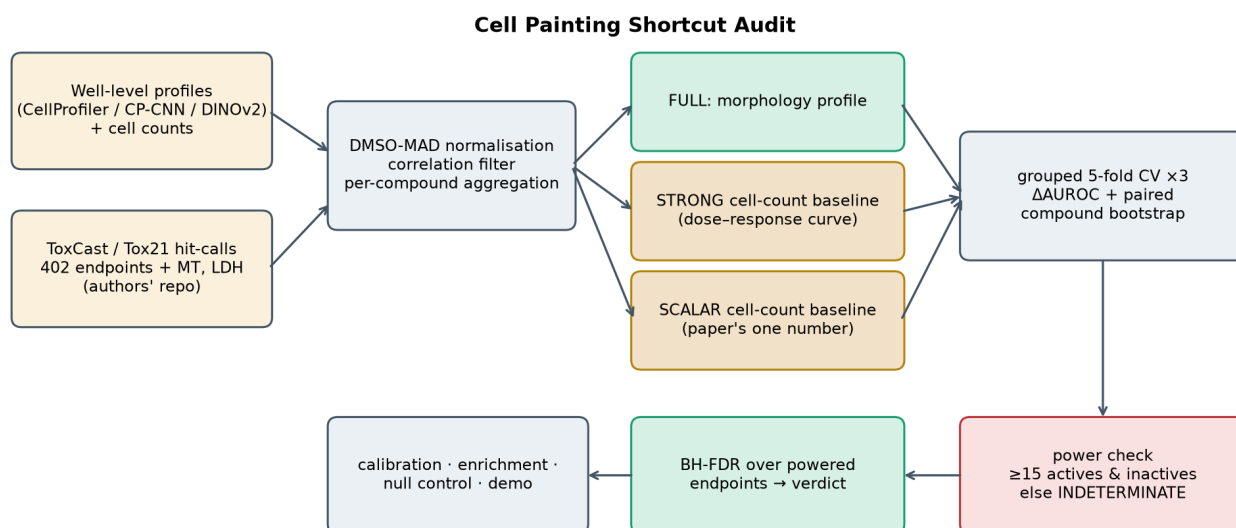
The main audit compares morphology *alone* with cell count *alone*. The question "does morphology add information beyond count?" is the nested one, so a fourth XGBoost model sees the morphology profile **plus** the cell-count curve features (*nested*) and is compared with the cell-count-only model on identical folds, with its own label-permutation null (sd_0 estimated from permutation runs at the same settings), calibrated p-values, BH over powered endpoints, and the verdicts `adds_beyond_count` / `no_detectable_addition` (`scripts/22_nested_audit.py`). A median-imputed, standardised, L2-regularised logistic regression on the morphology features, whose regularisation strength is chosen by an inner 3-fold CV inside each training fold, serves as a sanity comparator for the tree model.

3.9 Implementation

The package `src/cpsa/` mirrors the pipeline: `data/` (`load_profiles`, `preprocess`, `load_labels`, `build_dataset`, `technical`), `models/` (`train_eval`: grouped CV, XGBoost wrapper, paired bootstrap; `baseline`, `full_model`: feature-set definitions), `stats/` (`power_check`, `multiple_testing`, `null_calibration`, `calibration`), `biology/` (`enrichment`, `feature_families`), `viz/plots`, and `audit.py` (end-to-end orchestration with joblib parallelism over endpoints). Numbered scripts in `scripts/` are thin command-line wrappers; every figure and number in this report is regenerated from `results/` by scripts 07, 12 and 13. Design choices that serve reproducibility: one parquet per (representation, aggregation) cached in `data/processed/`; fixed seeds everywhere; MD5-verified downloads; no hidden state in notebooks (notebooks are exploration only); 69 unit tests including a grouped-CV leakage test (duplicated compounds with noise features must score at chance), a toy BH reference, an empirical-null recovery test and a reproduction of the paper's unshuffled fold protocol. The demo reads only precomputed artifacts.

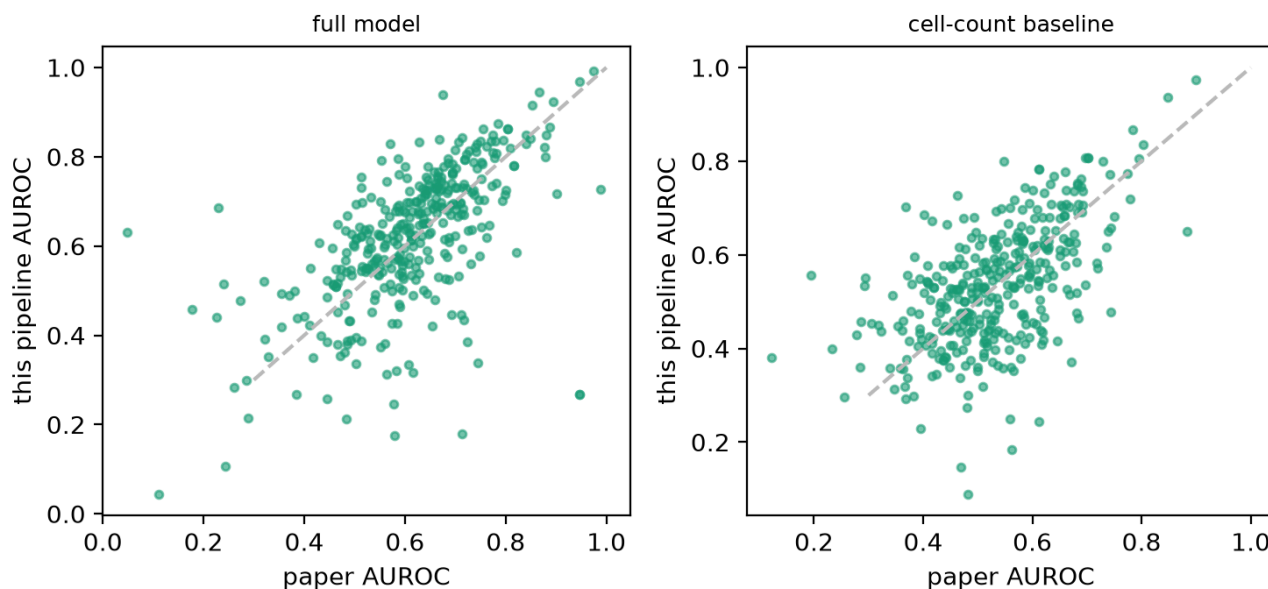
4. Results

Primary configuration (pre-specified in the project plan): **CellProfiler features**, **allpod aggregation**; numbers from `results/cellprofiler_allpod/` (`headline_numbers.json`, regenerated by `scripts/12_headline_numbers.py`). CP-CNN and DINOv2 are robustness checks and were audited with the same settings and their own matched permutation nulls. 404 endpoints were tested (292 cell-based, 72 cell-free, 38 cytotoxicity ToxCast/Tox21 endpoints plus the paper's MT and LDH hits); `cell_count` is the positive control. All three audits cover every endpoint (3 CV repeats, 1,000 bootstrap resamples), use the concentration-aware cell-count features, and were run once, in full, after the label-semantics corrections of §2.

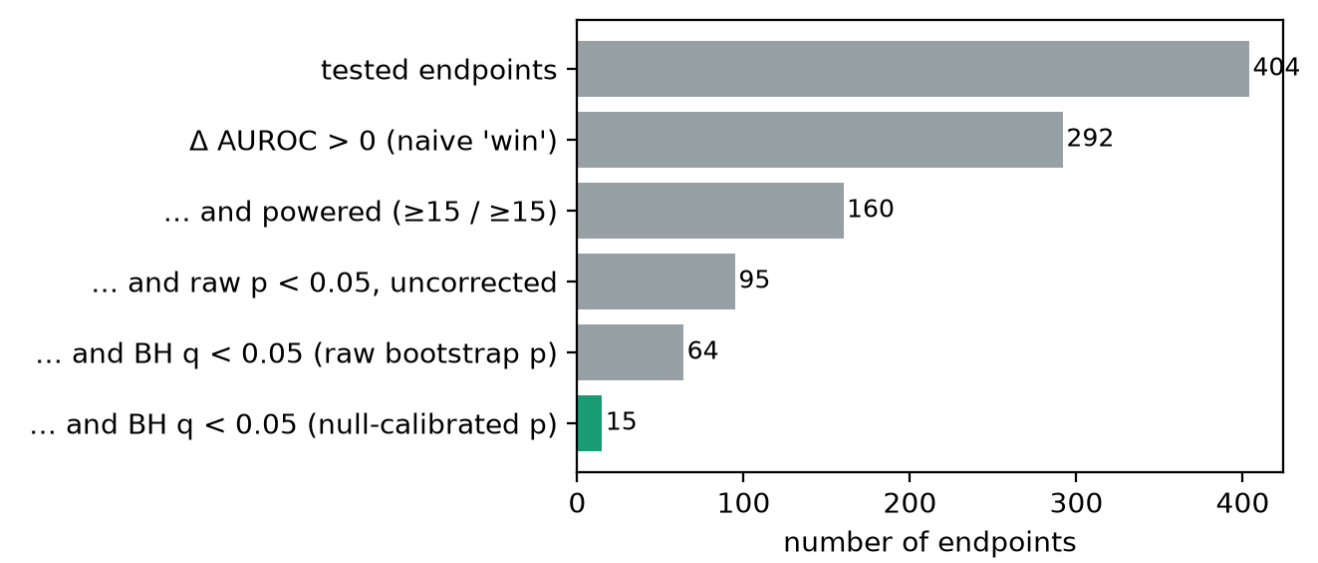


4.1 The pipeline reproduces the source paper's AUROC level

Against the paper's published per-endpoint AUROCs (CP-CNN features, `allpod`, 179 powered endpoints), re-running our models under the **paper's own protocol** (unshuffled `StratifiedKFold`, one run, the paper's row order; `scripts/11_paper_faithful_cv.py`) gives a mean AUROC difference of +0.004 (full model) and +0.007 (cell-count baseline), Pearson $r = 0.79 / 0.70$ across endpoints — limited by small-sample noise: for CellProfiler features and endpoints with ≥ 30 actives $r = 0.91 / 0.85$. Our default protocol (shuffled folds, 3 repeats averaged) scores the full model $\approx +0.04$ higher, i.e. the offset is a cross-validation-protocol effect, not a data-processing one.

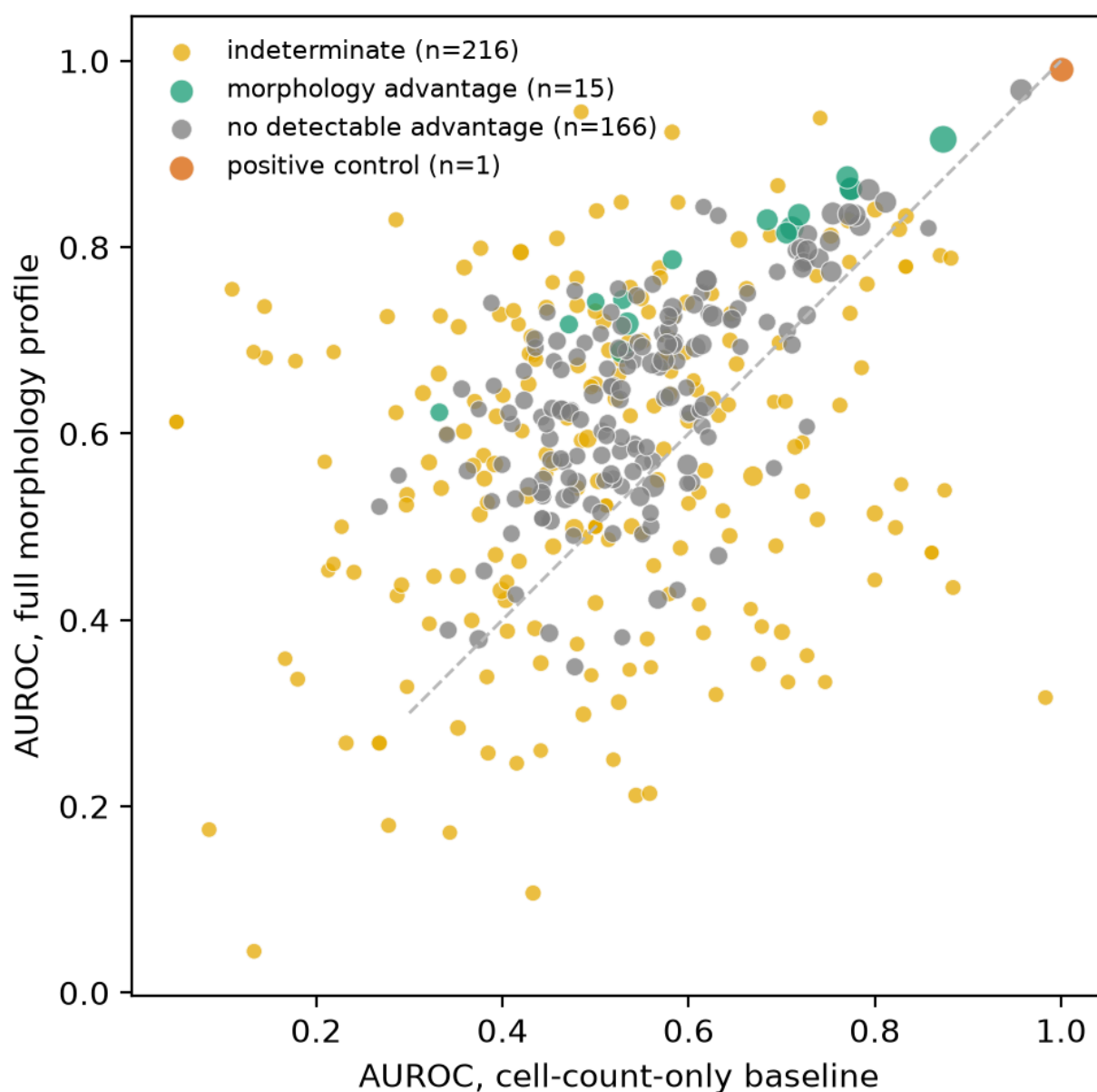


4.2 Most endpoints cannot support a claim, and naive wins are mostly not certifiable



step (CellProfiler, allpod)	endpoints
tested (excl. positive control)	404
Δ AUROC > 0 against the strong cell-count baseline — the naive "morphology wins"	292
... of which powered (≥ 15 actives and ≥ 15 non-hits); 223 (55 %) are indeterminate	160
... and raw bootstrap $p < 0.05$, uncorrected	95
... and BH $q < 0.05$ on raw bootstrap p -values (anti-conservative, §4.4)	64
... and BH $q < 0.05$ on null-calibrated p-values → morphology_advantage	15 (of 181 powered)

For indeterminate endpoints the reported AUROCs are essentially noise: the full-model AUROC has a standard deviation of 0.175 across indeterminate endpoints versus 0.117 across powered ones, and 56 of 223 lie outside [0.4, 0.8] (40 below 0.4). Reporting them as clean AUROCs would manufacture spurious wins and losses (figure below: indeterminate endpoints in yellow scatter from 0.02 to 1.0).



Yet effect sizes among the powered endpoints are broadly positive — $\Delta\text{AUROC} > 0$ for 160 of 181 (88 %), median $\Delta = +0.094$ (median AUROC 0.650 full vs 0.541 strong cell-count baseline) — so the limit is per-endpoint statistical certainty, not an absence of signal: with 15–60 actives per endpoint only large effects can be certified individually. **The certified count is a band, not a point:** it is 67, 38, 23, 15 and 5 endpoints if the null sd is taken as 1.0, 1.2, 1.33, 1.45 and 1.65 (our matched estimate is 1.37), so it must not be read as "how many endpoints benefit".

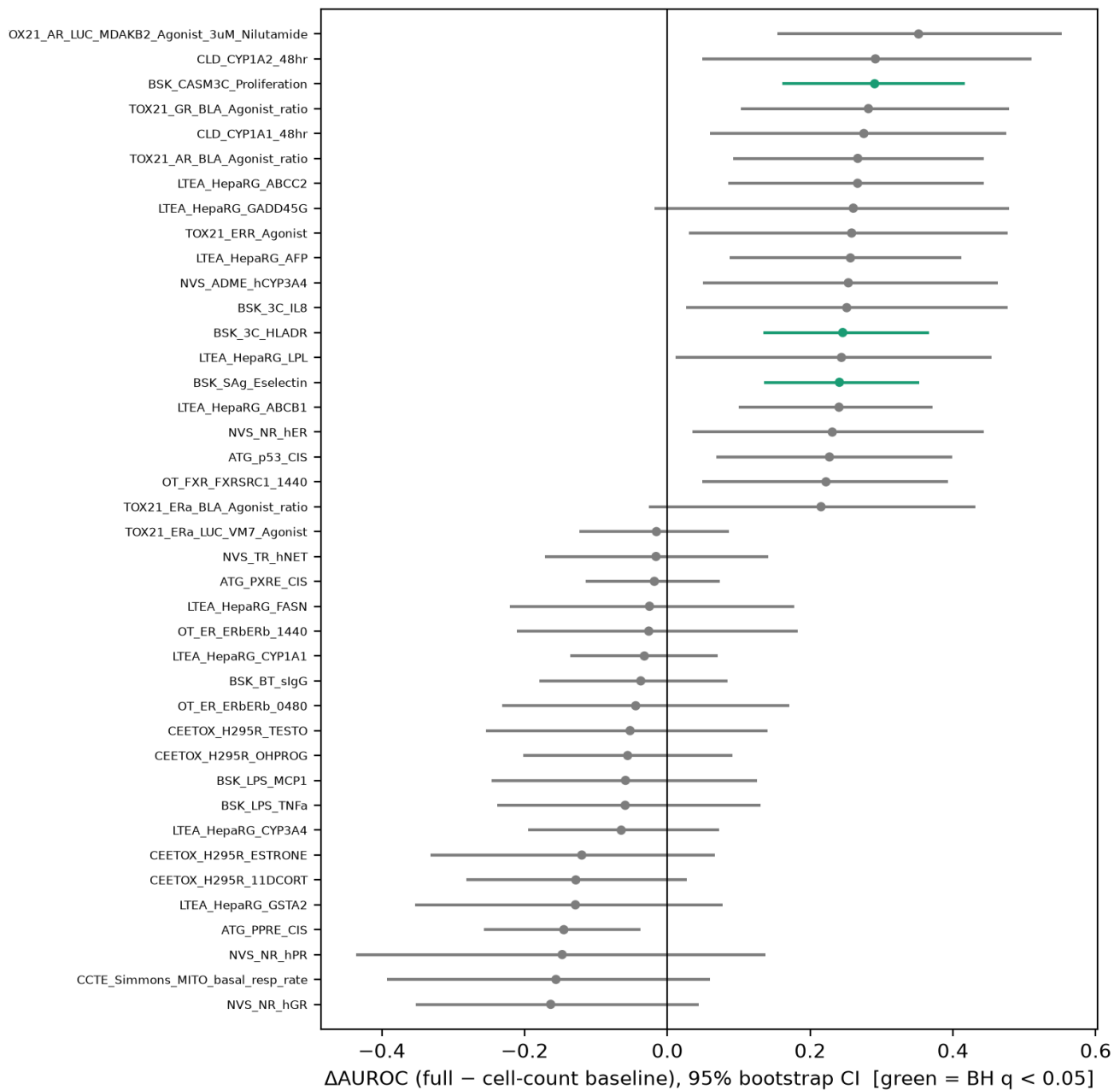
4.3 The certified endpoints overlap strongly across representations

representation (allpod)	null sd_0 (powered null runs)	certified vs strong cc	vs scalar cc	raw-bootstrap counts (strong / scalar)
CellProfiler (primary)	1.37 (379)	15	51	64 / 75
CP-CNN	1.54 (274)	15	29	60 / 84
DINOV2	1.27 (58; noisy)	54	59	92 / 85

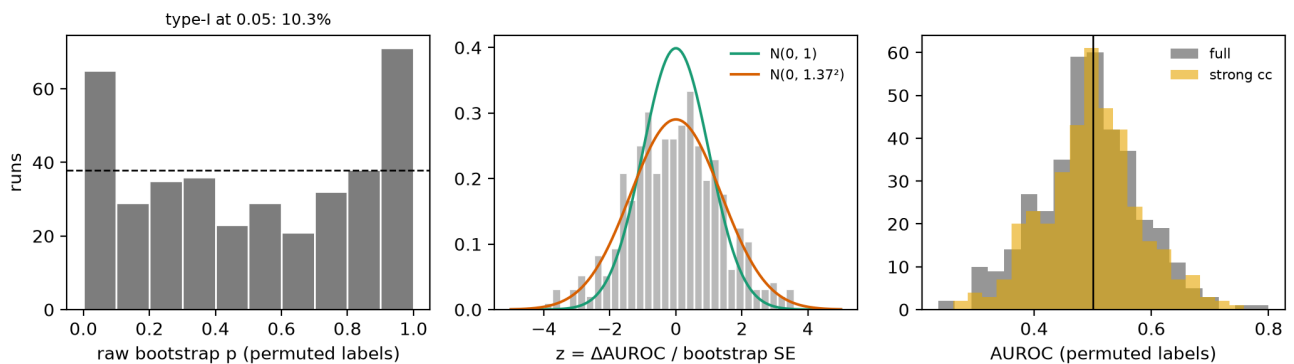
Every CellProfiler-certified endpoint and every CP-CNN-certified endpoint is also certified with DINOv2; **8 endpoints are certified under all three representations** (MT, the PR-bla and GR-bla antagonist assays, two BioMAP assays — a proliferation and an E-selectin readout — and three cytotoxicity endpoints: HEK293, ME-180, ERR-HEK293T); 22 are certified under at least two and 54 under any. Δ AUROC is rank-correlated across representations over the 181 powered endpoints (Spearman 0.70 CellProfiler vs CP-CNN, 0.66 vs DINOv2). The *number* certified depends strongly on the representation (15 / 15 / 54), and DINOv2's null rests on only 58 runs, so we report all three rather than a single headline.

The 15 endpoints certified with CellProfiler features:

endpoint	category	actives	AUROC strong cc → full	Δ	q CellProfiler	q CP- CNN	q DINOv2
T0X21_PR_BLA_Antagonist_ratio	cellbased	145	0.53 → 0.72	+0.184	0.002	0.001	0.000
cell_type__HEK293	cytotox	186	0.71 → 0.82	+0.110	0.002	0.007	0.001
tissue__kidney	cytotox	130	0.77 → 0.88	+0.105	0.002	0.054	0.000
cell_type__ME-180	cytotox	101	0.68 → 0.83	+0.145	0.002	0.006	0.000
T0X21_PXR_agonist	cellbased	124	0.53 → 0.69	+0.160	0.005	0.153	0.005
BSK_CASM3C_Proliferation	cellbased	34	0.33 → 0.62	+0.290	0.014	0.007	0.005
cell_type__ERR-HEK293T	cytotox	131	0.72 → 0.83	+0.116	0.014	0.024	0.000
tissue__intestinal	cytotox	100	0.71 → 0.82	+0.110	0.014	0.065	0.008
BSK_SAg_Eselectin	cellbased	30	0.50 → 0.74	+0.241	0.015	0.007	0.000
BSK_SAg_Proliferation	cellbased	56	0.58 → 0.79	+0.204	0.015	0.120	0.001
MT	native	377	0.87 → 0.92	+0.042	0.019	0.039	0.001
BSK_3C_HLADR	cellbased	39	0.47 → 0.72	+0.246	0.019	0.301	0.010
cell_type__HepG2	cytotox	150	0.77 → 0.86	+0.088	0.021	0.065	0.002
tissue__liver	cytotox	150	0.77 → 0.86	+0.088	0.021	0.065	0.002
T0X21_GR_BLA_Antagonist_ratio	cellbased	53	0.53 → 0.74	+0.215	0.029	0.021	0.000



4.4 Auditing the audit: raw bootstrap p-values are anti-conservative



Permuted-label runs use exactly the audit's settings (3 repeats, 1,000 resamples; an earlier null with fewer repeats and resamples had estimated sd 1.45 and produced a count of 9, which is why the count is now larger). In 379 powered CellProfiler runs (mean AUROC 0.49 full, 0.50 strong baseline, as they

should) the z-score has sd 1.37 where a valid test would give 1.0, so the raw p-value rejects at nominal 5 % in **10.3 %** of runs. Calibrating with an sd estimated from the first 190 runs (1.45) brings the error rate on the other 189 runs to **2.6 %** (nominal 5 %) and 0.5 % (nominal 1 %) against 7.9 % raw: conservative. For CP-CNN the inflation is larger (sd 1.54 from 274 runs; raw 17.9 %); the split-half calibrated rate is 6.5 % (nominal 5 %; 139 held-out runs, within sampling noise) and 0 % at nominal 1 %. For DINOv2 the null has only 58 powered runs (sd 1.27, raw 11.7 %); no split-half check is possible, and the uncalibrated pipeline made 2 BH calls in 60 permuted runs (0 for CellProfiler and CP-CNN). The null sd depends mildly on endpoint size for CellProfiler (1.44 / 1.35 / 1.21 for 15–29 / 30–59 / ≥ 60 actives); we use one pooled value per representation.

4.5 Weak baselines hide shortcuts

Against the paper's scalar cell-count baseline 51 endpoints are certified; 38 of them lose the advantage against the stronger cell-count baseline (2 are certified only against the strong baseline, where the many-feature model over-fits at small n). The strong baseline matters most for the native readouts:

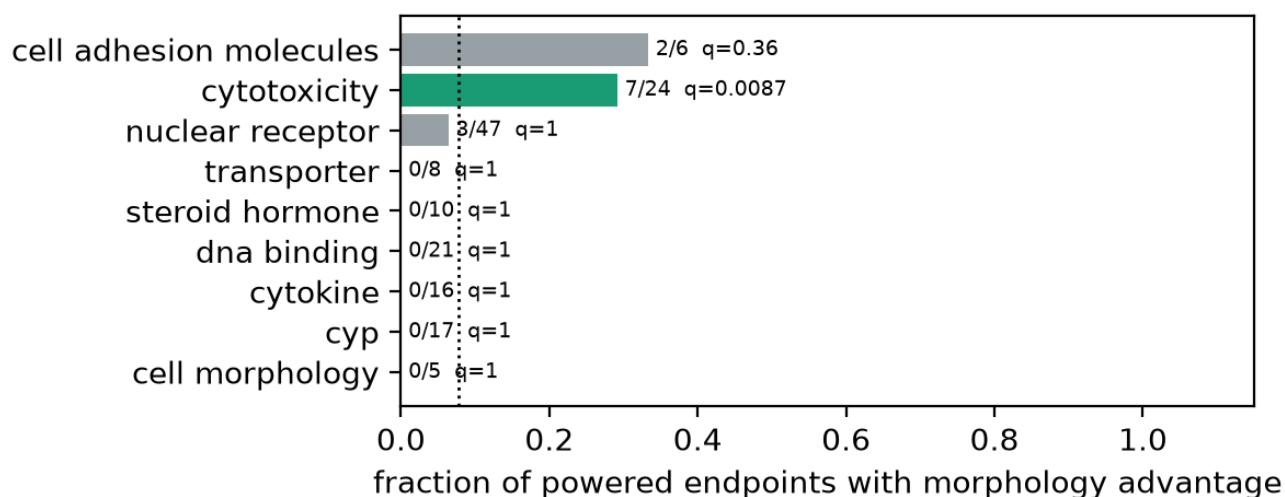
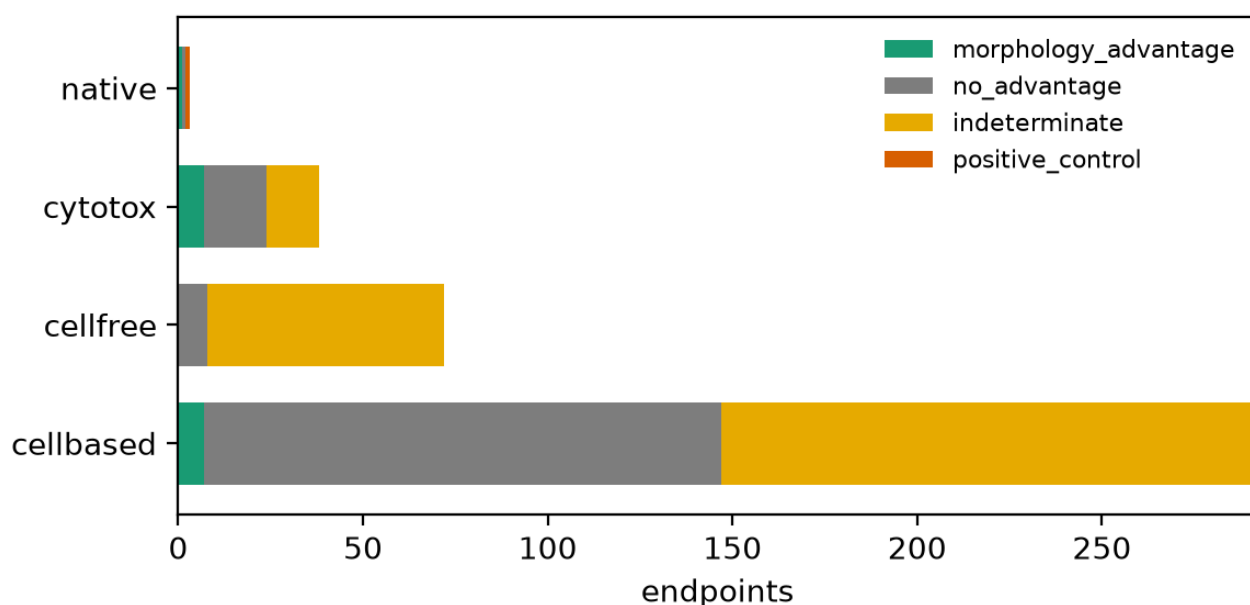
readout	scalar cc	strong cc	full (CellProfiler)	Δ vs strong	BH q (CP / CP-CNN / DINO)	verdict vs scalar
MT (metabolic activity)	0.835	0.873	0.916	+0.042	0.019 / 0.039 / 0.001	advantage (q < 0.001)
LDH (membrane damage)	0.936	0.957	0.969	+0.011	0.27 / 0.29 / 0.076	advantage (q = 0.016)
cell_count (positive control)	0.975	1.000	0.991	—	—	control passes

Morphology beats cell count for MT with all three representations but is **never** certified for LDH at the 5 % level (DINOv2 is borderline, q = 0.076) — in agreement with the paper — while the one-number baseline would have called LDH a win. The strong baseline reaching AUROC 1.000 on the cell-count positive control confirms that it captures everything cell count can give.

4.6 Where morphology wins

category	powered	certified (CellProfiler / CP-CNN / DINOv2)
cell-based reporter endpoints	147 / 292	7 / 10 / 36
cytotoxicity endpoints (cell-type / tissue summaries, 1 = cytotoxic)	24 / 38	7 / 4 / 17
cell-free (biochemical) endpoints	8 / 72	0 / 0 / 0
native (MT, LDH)	2	1 / 1 / 1

89 % of cell-free endpoints cannot be adjudicated and none of the 8 that can shows an advantage — consistent with the paper's finding that Cell Painting does not predict purely biochemical activity.



Enrichment — descriptive only. One-sided Fisher tests (BH within each grouping). Unadjusted, cytotoxicity endpoints look over-represented among certified endpoints with CellProfiler (7/24 vs 5 % elsewhere, odds ratio 7.7, $q = 0.003$) and DINOv2 (17/24 vs 24 %, OR 7.9, $q < 0.001$) but not with CP-CNN (4/24, $q = 0.36$). They are also far larger (median 97.5 actives vs 25 for cell-based endpoints), so they are simply easier to certify: restricted to endpoints with at least 50 actives the difference is not significant (CellProfiler cytotoxicity 7/22 vs cell-based 4/20, OR 1.87, $p = 0.49$; CP-CNN 4/22 vs 4/20, OR 0.89, $p = 1.0$; DINOv2 17/22 vs 10/20, OR 3.40, $p = 0.11$, undecided). We therefore report it as a pattern confounded with endpoint size, not as a finding. No other target family, cell line or tissue is enriched in more than one representation (DINOv2 flags cell-adhesion-molecule assays, 5/6, $q = 0.043$; CP-CNN flags antagonist-mode assays, $q = 0.073$; neither replicates). The "sub-lethal stress" interpretation rests on the compound-level analysis of §4.8, not on this enrichment.

Assay-level test. Endpoints from one assay are not independent, so each group also gets a permutation p-value in which whole assays (106 clusters) are the exchangeable unit (§3.7). The cytotoxicity enrichment against the strong baseline shrinks from Fisher $p = 0.001$ to $p = 0.12$ for CellProfiler (BH $q = 0.35$) and from 1×10^{-5} to $p = 0.039$ for DINOv2 ($q = 0.12$); against the paper's one-number baseline it is $p = 0.026$ (CellProfiler, $q = 0.08$). The cytotoxicity columns are themselves a handful of assay clusters

(`cell_type`, `tissue_`), so the data cannot support more than a descriptive reading, which is how we report it.

4.7 Not a technical shortcut

Plate number, mean well row/column, batch and replicate counts alone predict ToxCast labels at chance (mean AUROC 0.512; 11.6 % of endpoints above 0.6; `scripts/09_technical_probe.py`); adding them to the strong cell-count model changes nothing (0.563 vs 0.555), and for none of the 15 certified endpoints does "technical + cell count" match the full profile.

4.8 Compound level: where morphology adds information

Per compound, the *benefit* is how much better the full model ranks the compound (against the compounds of the opposite class) than the cell-count baseline does, averaged over the powered endpoints in which the compound is active. Compounds are grouped by the paper's own hit pattern across readouts (a mechanism-free grouping):

compound group	compounds	mean benefit [95 % CI]	benefit for all other compounds
active in Cell Painting only (no cell loss, MT or LDH hit)	133	+0.149 [0.103, 0.188]	+0.045
MT or LDH hit without cell loss	186	+0.128 [0.098, 0.162]	+0.041
cell loss (cell-count hit)	195	+0.010 [0.001, 0.019]	+0.087

Where cells die, counting them already ranks the compounds almost as well as morphology; where cells stay alive but change, morphology adds the information. (BH $q < 0.001$ for each group against the rest; the "inactive" group is not interpretable for active-compound benefit.) This is consistent with morphology reading sub-lethal stress before cells are lost, but it is an association within one dataset and has not yet been controlled for the baseline's own score range.

The `signal_class` labels split "no advantage" into *why*: of 181 powered endpoints, **30 are count-explained** (the cell-count baseline is already informative), **57 show morphology signal that cannot be certified as an advantage**, **79 show no detectable signal**, and 15 are certified. Gain importance by compartment, feature type and channel is nearly identical for certified and other endpoints (granularity 29.7 % of gain vs 24.5 % of features for certified, 30.2 % otherwise), so no single channel or compartment explains where morphology wins.

4.9 The cytotoxicity filter and the shortcut

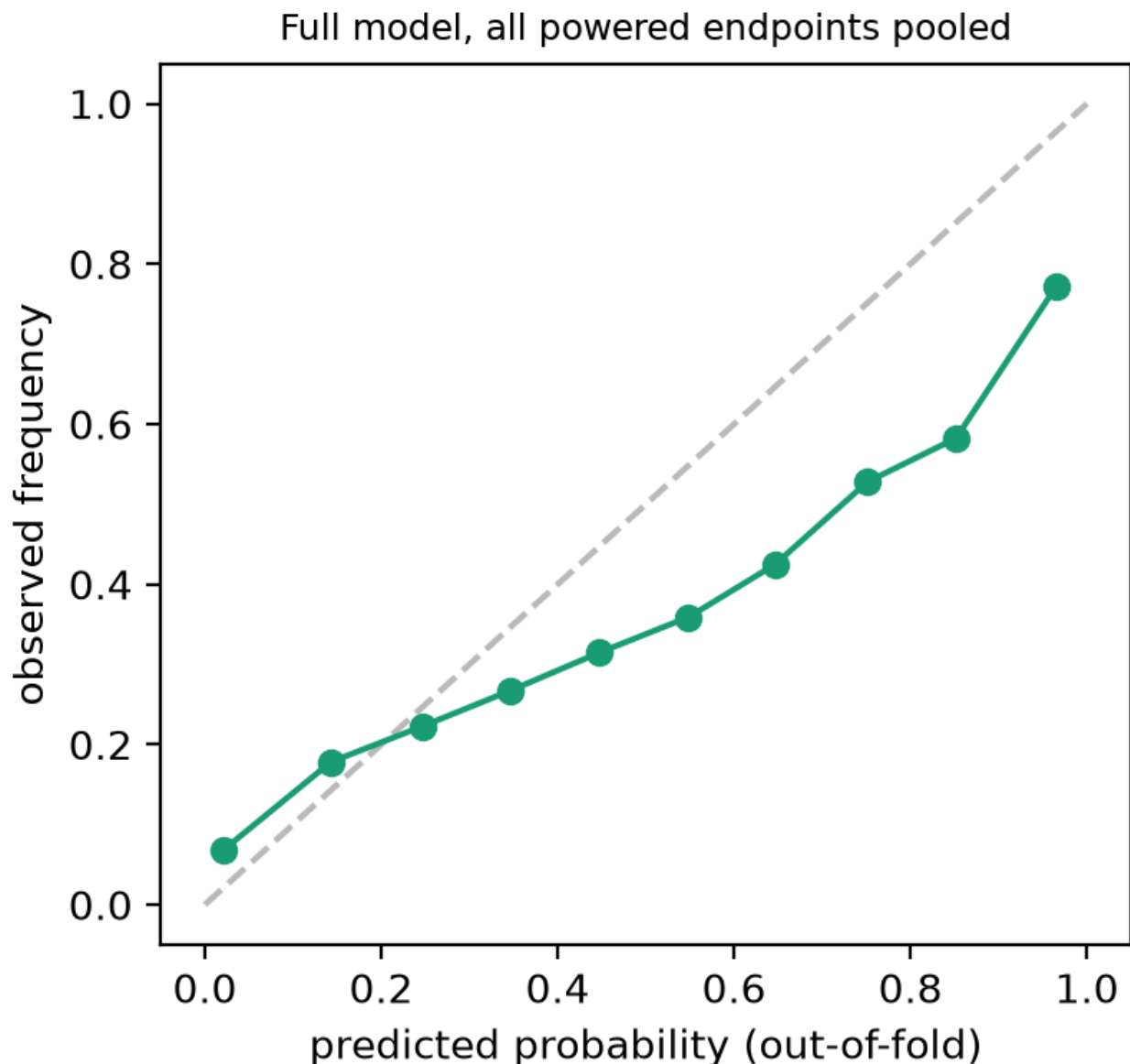
A ToxCast 0 is a "non-hit" (§3.3b), and the cytotoxicity filter that sets some hits to 0 applies only where a consensus cytotoxicity AC50 exists. Re-scoring the saved predictions by whether the filter could apply (`scripts/19_filter_stratified.py`, 48 cell-based endpoints scored in both strata): the filter could apply to 22.6 % of compound–endpoint pairs; the strong cell-count baseline is near chance in both strata (median AUROC 0.534 where it applies, 0.518 where it cannot), whereas the full profile is stronger where it cannot apply (0.649 vs 0.548) so morphology's median advantage is larger there (+0.118 vs +0.035). The strata are defined by compound properties (a matched cytotoxicity AC50 exists mainly for cytotoxic compounds), so this is descriptive, not causal.

4.10 Robustness

- **Power threshold** (min actives/non-hits per class 10 / 15 / 20 / 30): 19 / **15** / 22 / 26 certified endpoints out of 242 / 181 / 139 / 75 powered (a stricter threshold shrinks the BH pool and tends to raise the count; 15/15 was fixed in advance and not tuned). **$\alpha = 0.10$** : 36.
- **Chance-floored baseline** ($\Delta = \text{full} - \max(\text{baseline}, 0.5)$, so a below-chance baseline cannot inflate Δ): 13 of the 15 are retained.
- **Null sd** (§4.2): 67 / 38 / 23 / 15 / 5 certified at sd 1.0 / 1.2 / 1.33 / 1.45 / 1.65.
- **Aggregation rule** (CellProfiler, same settings and the same null): certified endpoints vs the strong baseline **all 18**, **allpod 15**, **allpodcc 3** (vs scalar: 56 / 51 / 60). **allpodcc** averages only wells below the cell-count point of departure, which by construction leaves little for any model to separate from the full cell-count curve, and MT is no longer certified there ($q = 0.90$); **all** certifies MT ($q = 0.004$). The aggregation dependence of the count is a limitation; **allpod** is the paper's default and our pre-specified primary.
- **Chemical-similarity folds** (Butina clusters of ECFP4 similarity ≥ 0.4 as CV groups, 921 clusters; `-group-by chem_cluster`; own permutation null, 88 powered runs, sd 1.44): median full-model AUROC 0.650 $\rightarrow$ 0.651 and median Δ AUROC 0.094 $\rightarrow$ 0.091 over the 181 powered endpoints (Spearman 0.83 between the two sets of Δ); 24 endpoints are certified and **all 15 of the primary set are among them**. Structural analogues in different folds therefore do not explain the result.
- **Representation**: §4.3.

4.11 Calibration

Raw probabilities are not trustworthy as confidences: for every one of the 181 powered endpoints the calibration slope is below 0.8 (median 0.33, i.e. strongly over-confident, a consequence of class re-weighting), the median expected calibration error is 0.081, and only 32 % of powered endpoints have a Brier score better than the prevalence predictor (median Brier skill -0.07). Cross-validated Platt recalibration lowers the median Brier score from 0.112 to 0.104. The demo therefore shows recalibrated probabilities together with each endpoint's calibration diagnostics.



4.12 Does morphology add information beyond cell count? Nested model, logistic comparator, detectable effect

The main audit compares morphology *alone* with cell count *alone*. The nested comparison (§3.8) gives the full model the cell-count curve as well and asks whether it then beats the cell-count-only model (CellProfiler `allpod`, 181 powered endpoints, identical folds):

- Morphology adds information on top of count.** The nested model's median AUROC is 0.661 against 0.541 for the cell-count-only model, a median gain of +0.091 (the alone-versus-alone gain is +0.094), and the gain is positive for 86 % of powered endpoints. With its own label-permutation null (600 runs, 555 powered; $sd_0 = 1.38$, close to the 1.37 of the main audit; raw bootstrap rejects 13 % of permuted endpoints at a nominal 5 %, and on held-out permutation runs the calibrated rate is 6.2 %) **23 endpoints are certified as adding information beyond count** (BH $q < 0.05$). All 15 endpoints certified in the main audit are among them, including MT ($q = 0.015$); LDH is still not certified ($\Delta = +0.016$, $q = 0.19$). The eight additional endpoints (five cell-based reporter assays, three cytotoxicity summaries) show information beyond the count curve once the model can combine both feature sets.

- **A linear comparator reaches a similar level.** A regularised logistic regression on the morphology features (regularisation chosen inside each training fold) has median AUROC 0.618 against 0.650 for the XGBoost full model, and is higher on 36 % of endpoints, so the endpoint-level signal is not an artefact of the tree model; the paper's XGBoost settings are retained because they score higher on most endpoints.
- **What each endpoint's data could detect.** The bootstrap interval of the AUROC difference, widened by sd_0 , gives a minimum detectable effect (80 % power, one-sided $\alpha = 0.05$; §3.5b). Its median over the 181 powered endpoints is 0.25 AUROC: only 18 powered endpoints could detect an advantage of 0.10, 36 of 0.15 and 54 of 0.20. Of the 15 certified endpoints 8 have a detectable effect of at most 0.10; 10 of the `no_advantage` verdicts rule out an advantage larger than 0.10 and 26 rule out one larger than 0.15. The remaining `no_advantage` verdicts mean "no detectable advantage" and say little about small effects, which is why the demo lists the detectable effect next to every verdict.

4.13 At organ-on-chip scale (simulation)

Liver-chip studies test tens of compounds, so we re-ran the audit on simulated data with known ground truth (the generator of `src/cpsa/simulate.py`; 20 to 200 compounds, four seeds per size, 720 endpoints per size, its own permutation null at each size; `scripts/24_chip_scale_study.py`, `results/chip_scale/`). No chip data is involved.

compounds	endpoints that cannot be judged (15/15 rule)	false credit on endpoints without an advantage, raw / calibrated p	power to credit a true advantage (calibrated)	smallest detectable advantage (AUROC)
20	100 %	0 % / 0 %	1 %	0.72
40	54 %	2 % / 0.5 %	11 %	0.50
60	29 %	4 % / 0.5 %	6 %	0.37
100	1 %	8 % / 1.5 %	33 %	0.27
200	0 %	8 % / 2 %	83 %	0.19

Three messages. (i) Calibration keeps false credit near zero at every size (the raw bootstrap reaches 8 % from 100 compounds on), so small designs do not produce spurious wins. (ii) The cost is power: with 20 to 60 compounds only very large advantages (AUROC difference of about 0.4 or more) could be certified, which is why chip results should be reported as the AUROC difference with its calibrated interval and detectable effect, not as a table of verdicts. (iii) Where compounds are run on several chips, the folds must be grouped by compound: with 40 compounds on three chips and random labels, cross-validation that treats each chip as independent reaches AUROC 0.85 (true value 0.5), against 0.51 when grouped by compound. A planning curve fitted on the 391 real plate endpoints (detectable effect $\approx e^{0.98 \cdot n_{\text{eff}}^{-0.54}}$, R^2 0.76, where n_{eff} is the balanced-design equivalent of the active and non-hit counts) agrees with the simulation within its band and drives the planner in the demo.

4.14 A second morphology source: EU-OPENSOURCE HepG2 (exploratory)

The EU-OPENSOURCE Bioactive Compound Set (Wolff et al.; Zenodo 10.5281/zenodo.19347244, CC BY 4.0) holds Cell Painting profiles of about 2,500 bioactives in HepG2 cells, imaged at four sites at 10 μM with four replicates. 327 of its compounds are also in our 967 profiled compounds (matched by the skeleton block of the InChIKey), so the existing ToxCast, Tox21, MT and LDH labels transfer, and 49 endpoints have at least 15 actives and 15 non-hits on that overlap (`src/cpsa/data/eu_os.py`,

`scripts/25_eu_os_replication.py, results/eu_os/`). HepG2 has one concentration, so only the paper-style single-number cell-count baseline exists (the median normalised object count of the compound's wells), not the dose–response baseline of the main audit. Two audits were run on identical compounds, endpoints and settings, each with its own label-permutation null (300 runs; sd_0 1.25 for HepG2 and 1.43 for the primary data; raw bootstrap rejects 10 % and 9 % of permuted endpoints at a nominal 5 %): HepG2 profiles, and the primary-hepatocyte CellProfiler profiles of the main audit.

	HepG2 profiles	primary hepatocytes
endpoints with a certified advantage (raw bootstrap would give)	16 (22)	1 (7)
median AUROC, full profile	0.71	0.68
median AUROC, cell-count baseline	0.57	0.61
MT: full profile / baseline / verdict	0.79 / 0.58 / certified	0.92 / 0.81 / certified
LDH	not certified ($q = 0.09$)	not certified

MT is certified and LDH is not in both systems, and 10 of the 16 HepG2 endpoints are also certified in the main audit against the strong baseline (MT, the PR-bla antagonist and PXR agonist assays, and the liver, kidney and intestinal cytotoxicity summaries). The 16 against 1 must not be read as HepG2 finding more morphology: the HepG2 cell count is one 10 μ M measurement from another cell system and is a much weaker baseline for labels defined in primary hepatocytes (for MT 0.58 against 0.81), and 327 compounds give much less power than the 967 of the main audit (the same scalar-baseline audit certifies 51 endpoints on all 967). It is the weak-baseline effect of §4.5 again: a verdict count depends on baseline strength and sample size. The rank agreement of per-endpoint AUROC between the two systems is moderate (Spearman 0.49) and that of the AUROC difference weak (0.13). We therefore report this as an exploratory second source that supports the MT result and the dataset-agnostic design of the code, not as a replication of the 15 certified endpoints.

5. Reliability analysis and limitations

What we did to make the audit itself trustworthy: label-permutation null with split-half validation (§4.4); a positive control that the strong baseline must solve (§4.5); a chance-floored baseline (§4.10); a technical-confound probe (§4.7); calibration diagnostics (§4.11); reproduction of the source paper's numbers under its own protocol (§4.1); a pre-specified power threshold with a sensitivity grid (§4.10); replication across three representations, aggregation rules and chemical-similarity folds (§4.3, §4.10); 50 unit tests covering preprocessing, grouped CV, bootstrap, BH, calibration, enrichment and the null calibration (`python -m pytest`).

Limitations, stated plainly:

- **Per-endpoint power is low.** With 15–60 actives per endpoint only large effects are individually certifiable (median minimum detectable effect 0.25 AUROC, §4.12); 55 % of endpoints are indeterminate, and the certified count (15 / 15 / 54 by representation; 5 to 67 for CellProfiler across plausible null sds) is a band limited by power, not an estimate of how many endpoints benefit.
- **Empirical-null calibration is approximate.** sd_0 is estimated from permutation runs at the audit's own settings (CellProfiler 379 powered runs, CP-CNN 274, DINOv2 only 58, so the DINOv2 calibration is the weakest and has no split-half check); it varies mildly with endpoint size (CellProfiler 1.44 / 1.35 / 1.21 for small / medium / large endpoints) and we use one pooled value per representation. The CP-CNN split-half error rate (6.5 % at nominal 5 %) shows the calibration

is not exactly conservative everywhere. A per-endpoint permutation test would be cleaner but costs $\sim 10^5$ model fits per configuration.

- **Dependence in BH.** Endpoints share compounds and assays and are correlated. BH controls FDR under independence and under positive dependence, which is the plausible structure here (the effective number of independent tests is smaller than the nominal count, so the correction is conservative), but positive dependence is assumed rather than verified. The enrichment p-values have an assay-level counterpart (§4.6).
- **"Morphology" is not orthogonal to cell density.** CellProfiler (neighbour counts, granularity, image-level features) and especially embedding features can encode confluence and clumping; the strong baseline sees only cell-count summaries. A "morphology advantage" therefore means *information beyond the cell-count curve*, which may include other density-related image properties.
- **Aggregation dependence.** The strong-baseline count varies with the aggregation rule (§4.10); `allpod` is the paper's default and our pre-specified primary.
- **Baseline strength has no ceiling.** A still stronger cell-count model (more features, tuning) might certify fewer endpoints; ours is a deliberate best-effort shortcut, not an upper bound.
- **What a 0 means.** In the ToxCast matrices a 0 is *not* "tested and negative": it is no hit, a hit overridden by the cytotoxicity filter, or a tie. The filter (hit set to 0 if the endpoint AC50 exceeds half the matched consensus cytotoxicity AC50) applies only where such a consensus exists (≥ 20 % of matched viability tests fired, ≤ 100 μM ; about 35 % of cell-based records), so many cell-based labels are not cytotoxicity-adjusted. Untested pairs stay missing (61.9 % of the cell-based matrix; pinned by a regression test). We say "non-hit" for the 0-class; the `n_inactive` column keeps its name. A stratified, descriptive check is in §4.9.
- **Irregular design.** Concentration series differ by source batch (952 compounds on 0.0456–100 μM , 14 on a 100 $\times$ lower series, one with 7 points) and replicate counts vary (829 single-well compound-concentration cells); dose-response baseline features therefore include concentration-interpolated cell counts and the technical probe includes well counts.
- **Label quality.** ToxCast hit-calls come from heterogeneous assays and cell systems; the 38 cytotoxicity columns are cell-type/tissue summaries where 1 = *cytotoxic* (the paper's 48 is before a minimum-5-positive / 5-negative filter); actives are rare for many endpoints; structurally related compounds in different folds could inflate all models equally.
- **Scope.** Single donor/pool of primary hepatocytes (donor variability not modeled), one 44 h time point, 2D plate assay — not an organ-on-chip; compound-level modeling means plate/well covariates are probed separately rather than included.

6. Motivation and impact

Regulatory context. FDA's March 2026 draft guidance "General Considerations for the Use of New Approach Methodologies in Drug Development" (Federal Register 2026-05390, 19 March 2026) names four factors for validating a NAM — context of use, human biological relevance, **technical characterization** and **fit for purpose**. An audit that tells a developer whether a profiling model's predictive power is a real signal or a trivial confound, endpoint by endpoint and with explicit uncertainty, is a concrete instance of technical characterization; indeterminate verdicts and calibration diagnostics speak to fit for purpose.

Scientific context. The Cell Painting community has documented that cell count alone can predict many small-molecule bioactivity benchmark labels (Seal et al., *Nat. Commun.* 2026), and the JUMP-CP resource (Chandrasekaran et al., *Nat. Methods* 2024) is the large public dataset where such baselines

matter. Ewald et al. (2026) apply that insight to hepatocyte cytotoxicity; we turn it into a reusable, statistically controlled procedure and show, with a permutation control, which of its most natural implementations (plain bootstrap p-values) would have over-called.

Organ-on-chip relevance. Chip assays will produce rich image-based readouts on small numbers of compounds — exactly the regime where multiplicity, low power and trivial survival confounds bite. The audit is dataset-agnostic: it needs profiles, a cell-count-like column and binary endpoints.

6.1 Extension to organ-on-chip: what carries over and what changes

No public liver-chip dataset combines images with compound-level toxicity labels, so this extension is a requirements analysis supported by the simulation of §4.13, not a chip result. The primary data here are 2D plate cultures at one time point.

Carries over. The question (does the full profile beat a cell-count baseline?); the paired compound bootstrap and label-permutation calibration; the power check and the indeterminate verdict; the detectable-effect report; the label hygiene of §3 (untested stays missing, a zero is a non-hit); grouped cross-validation.

Changes. (1) **Labels:** ToxCast and Tox21 will not cover most chip compounds, so endpoints become functional readouts (albumin, urea, CYP3A4 activity, ATP, LDH efflux) and clinical DILI categories (for example DILrank), a handful instead of about 400. (2) **Baseline:** the 2D cell count needs a 3D proxy (nuclei per z-stack, total DNA-channel signal, tissue area or volume), chosen in advance because it defines what the audit means. (3) **Features:** chips are imaged again (confocal z-stacks, PDMS autofluorescence, dye penetration) and profiled; embeddings trained on 2D monolayers should not be assumed to transfer. (4) **Grouping:** compound, chip, field and donor or lot, because replicate chips and shared donors otherwise leak (§4.13). (5) **Sample size:** typical chip studies fall in the 20 to 60 compound range of §4.13, where certification needs very large effects, so results are best reported as estimates with calibrated intervals, or pooled across studies or sites.

A realistic first pilot. Twenty to sixty compounds spanning the DILrank categories, two or three functional endpoints chosen in advance, a defined 3D cell-count proxy, folds grouped by compound and donor, and the result reported as the AUROC difference with its calibrated interval and its detectable effect. An open liver-chip dataset with compound labels does not exist yet; a chip-validation consortium could produce one, and the audit is meant to be run on it. The demo's Organ-on-chip tab provides the planner and the checklist.

7. Reproduction instructions

See README.md. Entry points: `scripts/01_download_zenodo_data.py`, `02_fetch_labels.py`, `03_run_audit.py`, `08_null_control.py`, `15_calibrate_verdicts.py`, `04_run_enrichment.py`, `22_nested_audit.py` (nested model, null and calibration), `23_detectable_effect.py`, `24_chip_scale_study.py` (chip-scale simulation and planning curve), `25_eu_os_replication.py` (HepG2 second source), `26_fetch_cell_line_images.py` (demo images), `07_finalize_results.py`, `12_headline_numbers.py`, `13_sensitivity_postprocess.py`; demo `streamlit run demo/app.py`. Everything is CPU-only with fixed seeds and MD5-verified downloads. Measured cost on a 10-core laptop (uncontended): CP-CNN full audit of all 404 endpoints $\approx$ 14 min, CellProfiler $\approx$ 29 min; 200 permutation runs $\approx$ 10–25 min. Unit tests: `python -m pytest`.

8. References

1. Ewald, J.D., Titterton, K.L., Bäuerle, A., et al. Cell Painting for cytotoxicity and mode-of-action analysis in primary human hepatocytes. *Cell Systems* 17(5), 101566 (2026). doi:10.1016/j.cels.2026.101566
2. Zenodo record 10.5281/zenodo.17067683 — Axiom OASIS profiles and metadata (CC-BY 4.0).
3. Authors' analysis repository, github.com/jessica-ewald/2024_09_09_Axiom_OASIS (BSD-3-Clause).
4. Seal, S., Dee, W., Shah, A., et al. Counting cells can accurately predict small-molecule bioactivity benchmarks. *Nat. Commun.* 17, 2436 (2026). doi:10.1038/s41467-026-68725-5
5. Chandrasekaran, S.N., Cimini, B.A., Goodale, A., et al. Three million images and morphological profiles of cells treated with matched chemical and genetic perturbations. *Nat. Methods* 21, 1114–1121 (2024). doi:10.1038/s41592-024-02241-6
6. U.S. FDA. General Considerations for the Use of New Approach Methodologies in Drug Development; Draft Guidance for Industry; Availability. *Federal Register*, 19 March 2026, document 2026-05390.
7. Benjamini, Y., Hochberg, Y. Controlling the false discovery rate: a practical and powerful approach to multiple testing. *J. R. Stat. Soc. B* 57, 289–300 (1995).
8. Efron, B. Large-scale simultaneous hypothesis testing: the choice of a null hypothesis. *J. Am. Stat. Assoc.* 99, 96–104 (2004).
9. Chen, T., Guestrin, C. XGBoost: a scalable tree boosting system. *KDD* (2016).
10. Cell Painting Gallery (cpg0037-oasis), AWS Registry of Open Data, CC0 — used only for a dozen demo images.

Appendix A. Output schema (`results/audit_table.csv`)

One row per endpoint: `endpoint_id`, annotations (`category`, `assay_target_family`, `assay_design_type`, `cell_short_name`, `tissue`, `endpoint_description`), `n_compounds`, `n_active`, `n_inactive`, `powered`, `scalar_cc_AUROC/PRAUC`, `strong_cc_AUROC/PRAUC`, `full_AUROC/PRAUC`, `delta_AUROC` (+ `delta_ci_lo/hi`, `delta_z`), `bootstrap_p` (raw), `calibrated_p`, `fdr_q`, `verdict`, `verdict_uncalibrated`, the `_scalar` counterparts versus the one-number baseline, and `calibration_brier`, `brier_skill`, `ece`, `calib_slope`, `brier_recalibrated`.

Appendix B. All configurations

`results/sensitivity_summary.csv`. `cellprofiler_allpod`, `cpcnn_allpod` and `dino_allpod` audit every endpoint; the other configurations (`CellProfiler_all`, `allpodcc`, and chemical-cluster folds `_chem`) were run on powered endpoints only. Each representation uses its own matched-settings permutation null; CellProfiler aggregation variants use the CellProfiler `allpod` null (assumed shared), and the chemical-cluster run its own.

configuration	endpoints modeled	powered	certified vs strong cc	certified vs scalar cc	raw-bootstrap (strong / scalar)	null sd ₀ used	median AUROC full / strong / scalar (powered)
cellprofiler_all	181	181	18	56	52 / 81	1.374	0.638 / 0.543 / 0.509
cellprofiler_allpod	404	181	15	51	64 / 75	1.374	0.650 / 0.541 / 0.540
cellprofiler_allpod_chem	181	181	24	51	56 / 71	1.442	0.651 / 0.531 / 0.540
cellprofiler_allpodcc	181	181	3	60	44 / 88	1.374	0.657 / 0.539 / 0.521
cpcnn_allpod	404	181	15	29	60 / 84	1.54	0.654 / 0.547 / 0.534
dino_allpod	404	181	54	59	92 / 85	1.273	0.680 / 0.544 / 0.557

Appendix C. Power-threshold and a grid (CellProfiler_allpod)

results/cellprofiler_allpod/sensitivity_postprocess.csv.

min actives & non-hits	α	baseline	p-values	powered	certified
10	0.05	strong_cc	null-calibrated	242	19
10	0.05	scalar_cc	null-calibrated	242	39
10	0.1	strong_cc	null-calibrated	242	38
10	0.1	scalar_cc	null-calibrated	242	76
15	0.05	strong_cc	null-calibrated	181	15
15	0.05	scalar_cc	null-calibrated	181	51
15	0.05	strong_cc	raw bootstrap	181	64
15	0.05	scalar_cc	raw bootstrap	181	75
15	0.1	strong_cc	null-calibrated	181	36
15	0.1	scalar_cc	null-calibrated	181	69
15	0.1	strong_cc	raw bootstrap	181	95
15	0.1	scalar_cc	raw bootstrap	181	94
20	0.05	strong_cc	null-calibrated	139	22
20	0.05	scalar_cc	null-calibrated	139	50
20	0.1	strong_cc	null-calibrated	139	34
20	0.1	scalar_cc	null-calibrated	139	64
30	0.05	strong_cc	null-calibrated	75	26
30	0.05	scalar_cc	null-calibrated	75	35
30	0.1	strong_cc	null-calibrated	75	39
30	0.1	scalar_cc	null-calibrated	75	44